

Problem 1

a) KVL:

$$V_L(s) = V_R - V_S = sL I_L(s) - L i_L(0)$$

KCL:

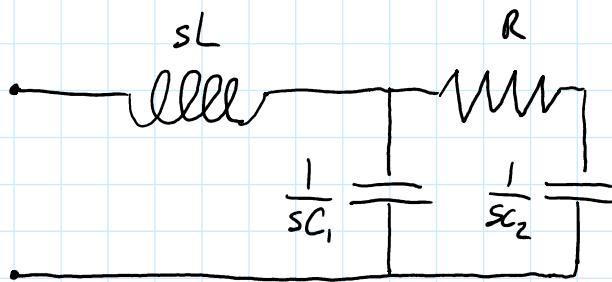
$$\begin{aligned} I_C(s) &= I_R - I_S = \frac{V_C(s)}{(sC)^{-1}} - C V_C(0) \\ &= sC V_C(s) - sV_C(0) \end{aligned}$$

$$b) Z(s) = \frac{V(s)}{I(s)}$$

$$V_L(s) = sL I_L(s) \Rightarrow Z_L(s) = \frac{V_L(s)}{I_L(s)} = sL$$

$$I_C(s) = sC V_C(s) \Rightarrow Z_C(s) = \frac{1}{sC}$$

c) Circuit in s-domain



$$Z(s) = \frac{\frac{1}{sC_1} \left(R + \frac{1}{sC_2} \right)}{R + \frac{1}{sC_1} + \frac{1}{sC_2}} + sL$$

$$= \frac{R + \frac{1}{sC_2}}{R sC_1 + 1 + \frac{sC_1}{sC_2}} + sL = \frac{1 + sRC_2}{s^2 RC_1 C_2 + sC_2 + sC_1} + sL$$

$$d) Z(j\omega) = \frac{1 + j\omega RC_2}{s^2 RC_1 C_2 + sC_2 + sC_1} = \frac{1 + j\omega \cdot 2,5 \cdot 10^3 \cdot 10 \cdot 10^{-6}}{s^2 RC_1 C_2 + sC_2 + sC_1}$$

$$d) Z(j\omega) = \frac{1 + j\omega RC_2}{(j\omega)^2 RC_1 C_2 + j\omega (C_1 + C_2)} = \frac{1 + j\omega \cdot 2,5 \cdot 10^{-3} \cdot 10 \cdot 10^{-6}}{-\omega^2 \cdot 2,5 \cdot 10^{-3} \cdot (10 \cdot 10^{-6})^2 + j\omega (20 \cdot 10^{-6})}$$

$$\rightsquigarrow = 2 \cdot 10^{-5} j\omega + \frac{4 \cdot 10^6 + 10^8 j\omega}{80 j\omega - \omega^2}$$

$$e) H(s) = \frac{V_2(s)}{V_1(s)}$$

$$V_2(s) = \frac{I(s)}{sC}$$

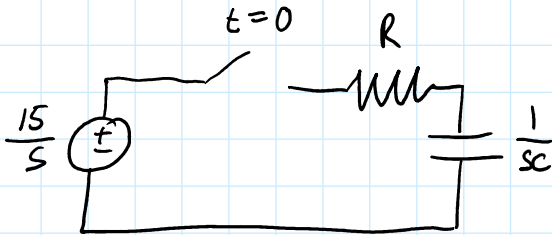
$$V_1(s) = Z(s) I(s) = I(s) \cdot \left(R + \frac{1}{sC}\right)$$

$$\frac{V_2(s)}{V_1(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{sCR + 1}$$

$$H(j\omega) = \frac{1}{RCj\omega + 1}$$

Problem 2

a) Laplace:



$$Z = R + \frac{1}{sC} \Rightarrow V_C(s) = \frac{V_S}{s} \left(\frac{\frac{1}{sC}}{R + \frac{1}{sC}} \right) = V_S \left(\frac{1}{s(RCs + 1)} \right)$$

$$\frac{1}{s(RCs + 1)} = \frac{A}{s} + \frac{B}{RCs + 1}$$

$$1 = A(RCs + 1) + Bs \Rightarrow A = 1$$

$$s = -\frac{1}{RC} \Rightarrow -\frac{1}{RC} B = 1 \Rightarrow B = -RC$$

$$V_c(s) = V_s \left(\frac{1}{s} - \frac{RC}{RCs + 1} \right)$$

$$V_s \cdot \mathcal{L}^{-1} \left\{ \frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right\} = V_s \left(1 - e^{-\frac{1}{RC}t} \right)$$

$$V_c(t) = 15 \left(1 - e^{-\frac{1}{2,5 \cdot 10^3 \cdot 100 \cdot 10^{-6}}t} \right) = 15(1 - e^{-4t})$$

Time constant of RC-circuit is

$$RC = 0,25s$$

$$b) I_c(s) = sC V_c(s) - C V_c(0)$$

$$KVL: \frac{V_s}{s} = R I_c + V_c$$

$$= \frac{V_s}{s} = R(sC V_c - C V_c(0)) + V_c$$

$$\Rightarrow V_c = \left(\frac{V_s}{s} + RC V_c(0) \right) \cdot \frac{1}{RCs + 1}$$

$$= \frac{V_s}{s(RCs + 1)} + \frac{RC V_c(0)}{RCs + 1}$$

$$\Rightarrow \mathcal{L}^{-1} \{ V_c \} = 15(1 - e^{-4t}) + 5e^{-4t}$$

$$= 15 - 10e^{-4t}$$

$$c) \begin{array}{c} V_c \\ | \end{array}$$



Problem 3

a) $I = \frac{V_s}{Z}$, $Z = sL + R$

$$V_s = V_0 \mathcal{L}\{\cos(\omega_0 t)\} = V_0 \frac{s}{s^2 + \omega_0^2}$$

$$I(s) = \frac{sV_0}{s^2 + \omega_0^2} \cdot \frac{1}{sL + R}$$

b) Optional

$$c) I(s) = \frac{As + B}{s^2 + \omega_0^2} + \frac{C}{s + \frac{R}{L}} = A \frac{s}{s^2 + \omega_0^2} + \frac{B}{\omega_0} \frac{\omega_0}{s^2 + \omega_0^2} + \frac{C}{s + \frac{R}{L}}$$

$$\mathcal{L}^{-1}\{I(s)\} = A \cos(\omega_0 t) + \frac{B}{\omega_0} \sin(\omega_0 t) + C e^{-\frac{R}{L}t}$$

$A = 0,0159$, $\frac{B}{\omega_0} = 0,0150$, $C = -0,0159$, $\frac{R}{L} = 333$

d) $C e^{-\frac{R}{L}t}$ will die out as $t \rightarrow \infty$

Stationary part: $0,0159 \cos(\omega_0 t) + 0,0150 \sin(\omega_0 t)$

Identity: $\sqrt{a^2 + b^2} \cos(\theta + \varphi) = a \cos \theta - b \sin \theta$

$$\sqrt{a^2 + b^2} = 0,0218433, \varphi = -43,33^\circ \quad \varphi = \tan^{-1}\left(\frac{b}{a}\right)$$

$$i_o(t) = 0,0218 \cos(\omega_o t - 43,33^\circ)$$

$$e) H(s) = \frac{I(s)}{V(s)}$$

$$V_s = \frac{sV_o}{s^2 + \omega^2}, \quad I(s) = \frac{sV_o}{s^2 + \omega^2} \cdot \frac{1}{sL + R}$$

$$H(s) = \frac{\frac{sV_o}{s^2 + \omega^2} \cdot \frac{1}{sL + R}}{\frac{sV_o}{s^2 + \omega^2}} = \frac{1}{sL + R}$$

$$H(j\omega_o) = \frac{1}{j\omega_o L + R} = \frac{1}{100\pi j + 333}$$

$$|H(j\omega_o)| = \frac{1}{\sqrt{(100\pi)^2 + 333^2}} = 0,00218433807$$

$$H(j\omega_o) = \frac{333 - \omega_o j}{333^2 + \omega_o^2} = \frac{333}{333^2 + \omega_o^2} - \frac{\omega_o}{333^2 + \omega_o^2} j$$

$$\angle H(j\omega_o) = \arctan^{-1}\left(\frac{-\omega_o}{333}\right) = -43,33^\circ$$

f) Ratio of amplitudes:

$$\frac{0,0218}{10} = 0,00218 = |H(j\omega_o)|$$

$$\text{Phase: } -43,33^\circ = \angle H(j\omega_o)$$

Problem 4

$$a) \theta_m(t) = \int_0^t \omega_m(\alpha) d\alpha$$

Laplace:

$$\Theta_m(s) = \frac{\Omega_m(s)}{s} = \frac{K V_m(s)}{(\tau s + 1)s}$$

$$\Rightarrow H_2(s) = \frac{\Theta_m(s)}{V_m(s)} = \frac{K}{s(\tau s + 1)}$$

$$b) V_m(t) = u(t) \Rightarrow V_m(s) = \frac{1}{s}$$

$$\frac{\Omega_m(s)}{V_m(s)} = \frac{K}{\tau s + 1} \Rightarrow \Omega_m(s) = \frac{K}{s(\tau s + 1)}$$

$$\frac{1}{s(\tau s + 1)} = \frac{A}{s} + \frac{B}{\tau s + 1}$$

$$A(\tau s + 1) + Bs = 1 \quad s=0 \Rightarrow A=1$$

$$s = -\frac{1}{\tau} \Rightarrow -\frac{1}{\tau} B = 1 \Rightarrow B = -\tau$$

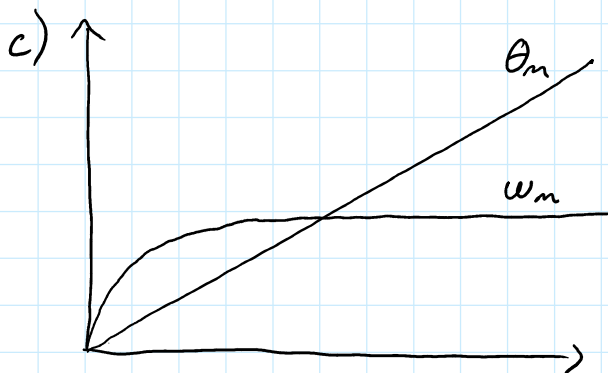
$$\Omega_m(s) = K \left(\frac{1}{s} - \frac{\tau}{\tau s + 1} \right)$$

$$\omega_m(t) = \mathcal{L}^{-1}(\Omega_m(s)) = K \left(1 - e^{-\frac{1}{\tau}t} \right)$$

$$= 23 - 23e^{-7,69t}$$

$$\theta_m(t) = \int_0^t \omega_m(\alpha) d\alpha = 23t + \frac{23}{7,69} e^{-7,69t}$$

$$= 23t + 2,99e^{-7,69t}$$



Assignment 2.1

$$a) V_o = L^{-1} \{ H(s) \cdot V_g \} = L^{-1} \left\{ \frac{50}{s^2} \cdot \frac{1000(s+5000)}{s^2 + 6000s + 25 \cdot 10^6} \right\}$$
$$= L^{-1} \left\{ \frac{As+B}{s^2} + \frac{Cs+D}{s^2 + 6000s + 25 \cdot 10^6} \right\}$$

$$\Rightarrow A = -\frac{1}{2500}, \quad B = 10, \quad C = \frac{1}{2500}, \quad D = -7.6$$

$$L^{-1} \left\{ -\frac{\frac{1}{2500}}{s} + \frac{10}{s^2} + \frac{1}{2500} \left[\frac{s+3000}{(s+3000)^2 + 4000^2} - \frac{5.5 \cdot 4000}{(s+3000)^2 + 4000^2} \right] \right\}$$

$$= -\frac{1}{2500} u(t) + 10t + \frac{1}{2500} e^{-3000t} \cos(4000t) - \frac{11}{5000} e^{-3000t} \sin(4000t)$$

b) Transient component is

$$\frac{1}{2500} e^{-3000t} \cos(4000t) - \frac{11}{5000} e^{-3000t} \sin(4000t)$$

c) Steady state component :

$$10t - \frac{1}{2500} u(t)$$